Beatriz Bastos Borges Nº03 CTII 350

Tarefa Básica

1. a)

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 5 \end{bmatrix} \quad \begin{matrix} -3 \\ +10 \end{matrix}$$

$\det A = 10 - 3$

$\boxed{\det A = 7}$

b)

$$B = \begin{bmatrix} -2 & -4 \\ 3 & 6 \end{bmatrix} \quad \begin{matrix} -(-12) \\ +(-12) \end{matrix}$$

$\det B = (-12) + 12$

$\boxed{\det B = 0}$

c)

$$C = \left[\begin{array}{ccc|cc} 3 & -1 & 1 & 3 & -1 \\ 2 & 1 & -1 & 2 & 1 \\ 1 & 4 & -2 & 1 & 4 \end{array}\right] \quad \begin{matrix} -1 & -(-12) & -4 \\ +(-6) & +1 & +8 \end{matrix}$$

$\det C = (-6) + 1 + 8 - 1 + 12 - 4 = 21 - 1$

$\boxed{\det C = 10}$

d)

$$D = \left[\begin{array}{ccc|cc} 3 & 2 & -1 & 3 & 2 \\ 2 & 3 & 1 & 2 & 3 \\ 1 & 1 & 4 & 1 & 1 \end{array}\right] \quad \begin{matrix} -(-3) & -3 & -16 \\ +36 & +2 & +(-2) \end{matrix}$$

$\det D = 36 + 2 - 2 + 3 - 3 - 16 = 41 - 21$

$\boxed{\det D = 20}$

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2. $A = (a_{ij})_{3\times3}$

$$a_{ij} \begin{cases} -3, \text{ se } i = j \\ 0, \text{ se } i \neq j \end{cases}$$

$$A = \begin{bmatrix} -3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{bmatrix}$$

$\det A = 0 + 0 + 0 - 27 - 0 - 0$

$\boxed{\det A = -27}$

$\boxed{\text{Alternativa A}}$

3. $\begin{vmatrix} x & 1 & x & x & 1 \\ 3 & x & 4 & 3 & x \\ 1 & 3 & 3 & 1 & 3 \end{vmatrix}$

$3x^2 + 4 + 9x - 9x - 1x^2 - 12x - 9 = -3$

$2x^2 - 3x - 5 = -3$

$2x^2 - 3x - 2 = 0$ $\quad a = 2 \quad b = (-3) \quad c = (-2)$

$\Delta = -3^2 - 4 \cdot 2 \cdot (-2)$

$\Delta = 9 + 16$

$\Delta = 25$

$$\frac{3 \pm \sqrt{25}}{2 \cdot 2} = \frac{3 \pm 5}{4} \begin{cases} x' = \frac{8}{4} = 2 \\ x'' = \frac{-2}{4} = \frac{-1}{2} \end{cases}$$

$\boxed{\text{Alternativa C}}$

4. $\begin{vmatrix} x-1 & -1 & 0 \\ 0 & x+1 & -1 \\ 2 & -1 & x+1 \end{vmatrix} \begin{matrix} x-1 & -1 \\ 0 & x+1 \\ 2 & -1 \end{matrix}$

$(x-1)(x+1)^2 + 2 + 0 - 0 - (x-1) - 0 = 2$

$(x-1)(x^2+2x+1) + 2 - x + 1 = 2$

$x^3 + 2x + x - x^2 - 2x - 1 + 2 - x + 1 = 2$

$x^3 + x^2 - 2x + 2 = 2$

$x^3 + x^2 - 2x = 0 \qquad a = 1 \quad b = 1 \quad c = (-2) \quad d = 0$

$x' + x'' + x''' = \frac{-b}{a} \rightarrow x' + x'' + x''' = \frac{-1}{1} = -1$

$\boxed{\text{Alternativa C}}$

5. $A = (a_{ij})_{3\times 2} \qquad a_{ij} = 2i - 3j$

$a_{11} = 2 - 3 = -1 \qquad a_{12} = 2 - 6 = -4$

$a_{21} = 4 - 3 = 1 \qquad a_{22} = 4 - 6 = -2$

$a_{31} = 6 - 3 = 3 \qquad a_{32} = 6 - 6 = 0$

$B = (b_{jk})_{2\times 3} \qquad b_{jk} = k - j$

$b_{11} = 1 - 1 = 0 \qquad b_{12} = 2 - 1 = 1 \qquad b_{13} = 3 - 1 = 2$

$b_{21} = 1 - 2 = -1 \qquad b_{22} = 2 - 2 = 0 \qquad b_{23} = 3 - 2 = 1$

$$A.B = \begin{bmatrix} -1 & -4 \\ 1 & -2 \\ 3 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 & -1 & -6 \\ 2 & 1 & 0 \\ 0 & 3 & 6 \end{bmatrix}$$

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$$\begin{bmatrix} 4 & -1 & -6 \\ 2 & 1 & 0 \\ 0 & 3 & 6 \end{bmatrix} \begin{matrix} 4 & -1 \\ 2 & 1 \\ 0 & 3 \end{matrix}$$

$det = 24 + 0 - 36 - 0 - 0 + 12$

$det = 36 - 36$

$det = 0$

$\boxed{\text{Alternativa C}}$

6.

$$A \cdot B = \begin{bmatrix} 2 & 0 & -1 \\ -1 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & -1 \\ -1 & 1 \\ 0 & 2 \end{bmatrix}$$

$$C = \begin{bmatrix} 2 & -4 \\ -2 & 2 \end{bmatrix} = 4 - 8 \qquad \boxed{det\, C = -4}$$

$\boxed{\text{Alternativa D}}$